Production Test Fixtures

Bed-of-nails • FCT jigs • Burn-in racks — building the production test station

Without a fixture, you can't reliably ship at volume

Audience: Engineers transitioning prototype to production

ICT • FCT • Boundary-scan • Burn-in • Cost analysis

Why Build Fixtures

Repeatability

Same test on every board.
Human variation eliminated.

Speed

Seconds per board
vs minutes with bench tools.

Coverage

Every test point verified.
90%+ net coverage possible.

Quality data

Pass/fail logged per board.
Serial number tracking.

Cost

Fixture amortized over 1000s
of boards = pennies/board.

Safety

Operator doesn't touch
live board.

ICT Bed-of-Nails

- Aluminium plate + vertical pogo pins matching every test point
- Pneumatic press lowers DUT (Device Under Test) onto pins
- Pins make contact via spring force (~50-100 g per pin)
- Tests: continuity, voltage, resistance per net
- Cycle time: 5-30 seconds per board
- Cost: ₹50-200k for custom fixture (per board design)
- Best for: medium-to-high volume production (≥ 1000 pcs)
- Tools: Genrad, Keysight, Acculogic, Teradyne

Building an ICT Fixture

- Step 1: Extract test point positions from PCB design (KiCad → custom script)
- Step 2: Map each test point to: net name + expected voltage/resistance
- Step 3: Design fixture plate (FR-4 or aluminum) with holes for pogo pin positions
- Step 4: Mount pogo pins in holes — solder wire from pin shaft to test electronics
- Step 5: Design alignment features for board placement (pins, slots)
- Step 6: Connect fixture to ICT tester (PXI, GPIB, custom microcontroller)
- Step 7: Write test program — power, sequence, measure, compare to expected
- Step 8: Validate on golden board + known-bad boards

FCT — Functional Circuit Test

- Powered-up test — board runs production firmware in test mode
- Connects via debug header (USB, UART, pogo) — not bed of nails
- Tests: comms, sensor reads, LED states, calibration values
- Pass/fail per board within seconds-to-minutes
- Calibrate sensors during FCT (write trim values to EEPROM)
- FCT often combined with ICT — ICT first, then FCT on passing boards
- FCT firmware ≠ production firmware (test routines + diagnostics added)
- After FCT pass → write production firmware + serial number

FCT Fixture Design

- Standoffs or pogo-pin jig that mounts the board
- Test connector header on the board interfaces to fixture
- Fixture has: power supply, USB-to-UART, JTAG/SWD programmer
- Microcontroller or PC runs the test sequence
- Result display: LED (red/green) + log file + barcode print
- Cost: ₹20-100k typical (less than ICT)
- Iteration friendly: easier to update than rebuilding ICT bed

JTAG Boundary Scan

- JTAG = standard chip-level test interface (boundary-scan)
- Tests interconnections between JTAG-capable ICs
- Coverage: 70-90% of nets if all chips support boundary scan
- Eliminates need for test point on every net (just need JTAG header)
- Fixture: simple 10-20 pin JTAG header + JTAG tester (XJTAG, ASSET InterTech)
- Cost: ₹10-30k for software + hardware (vs ₹100k+ for full ICT)
- Best for: dense BGAs where bed-of-nails can't reach
- Combine with FCT for full coverage

Burn-In Test

- Run boards at elevated temperature (60-85 °C) for 24-72 hours
- Accelerates infant-mortality failures (Arrhenius)
- Catches: bad solder joints, weak components, dry electrolytics
- Cost: thermal chamber + rack ~₹50k-2L
- Per-board cost: minutes of chamber time + power
- Industry standard: HASS (Highly Accelerated Stress Screen) follows HALT (testing)
- Used in: high-reliability products (medical, automotive, aerospace)
- Skipping burn-in → infant failures in field

Fixture Cost Analysis

Test type	Fixture cost	Per-board test cost	Notes
Manual bench	₹0 (existing)	5-30 min/board	OK for proto only
Flying probe	Lab cost ₹50-200/board	2-5 min/board	No fixture, slow
ICT bed-of-nails	₹50-200k	5-30 sec/board	Standard production
FCT	₹20-100k	30-120 sec/board	Tests functionality
JTAG boundary	₹10-30k SW + HW	5-10 sec/board	BGA-friendly
Burn-in chamber	₹50k-2L + rack	24-72 hr (parallel)	High-rel only

Fixture Build Workflow

- 1. Define test plan — what gets tested? Per board, per IC, per subsystem
- 2. Reserve test points / programming headers in PCB design (early)
- 3. Extract test point list + expected values from design
- 4. Buy/build fixture — pogo pins, frame, controller
- 5. Write test software (Python + Pyvisa for instrument control common)
- 6. Validate fixture with known-good + known-bad boards
- 7. Operator training + start-of-line calibration
- 8. Run for batch: log every result + alert on drift

Fixture Issues

- Test point spacing too tight — pogo pins damage each other
- Wear on pogo pins (1000s of insertions) → false failures
- Fixture grounding inadequate → noisy measurements
- Mechanical alignment loose → pins miss test points
- Software bug → false passes (worse than false fails)
- Operator skips broken fixture → defective boards ship
- Fixture design assumes only one PCB rev — doesn't accommodate Rev B

Vendor + Build Resources

- ICT vendors: Genrad (Teradyne), Keysight, Acculogic (Toronto, India ops)
- Pogo pin suppliers: Mill-Max, ECT Inc., Yamaichi (per the pin spec)
- Boundary scan: XJTAG, ASSET InterTech
- Indian fixture builders: contact local CMs for recommendations
- DIY fixture: 3D-printed plate + pogo pins from Amazon (low-volume only)
- Specialty: medical / mil-spec fixtures = custom procurement
- Cost: get 3+ quotes — fixture quote can vary 2× between vendors

Key Takeaways

Key takeaways from this lecture

Fixture mandatory at volume

No fixture = can't ship reliably.

ICT + FCT combo

Most common production setup.

Test points in design

Reserve early — can't add later.

Pogo pins wear

Replace every 10-50k insertions.

Validate fixture

With golden + known-bad boards.

Log every result

Trace defects back to root cause.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- Genrad / Keysight / Acculogic — ICT tester vendors
- XJTAG / ASSET InterTech — boundary scan
- Mill-Max / ECT — pogo pin suppliers
- IPC-9252 — electrical testing of unpopulated boards
- Phil's Lab YouTube — test fixture build videos
- Local CM partnerships — fixture build services
- Pyvisa / pyVISA-py — Python instrument control

Thank you • Build something this week.